

# Multisensory Integration in XR

HCPM Workshop 2026



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# Four words that get mixed up

## **multisensory**

several senses are stimulated at once. A property of a stimulus, a scene or a system

## **crossmodal**

information in one modality **changes** perception, attention or judgement in another

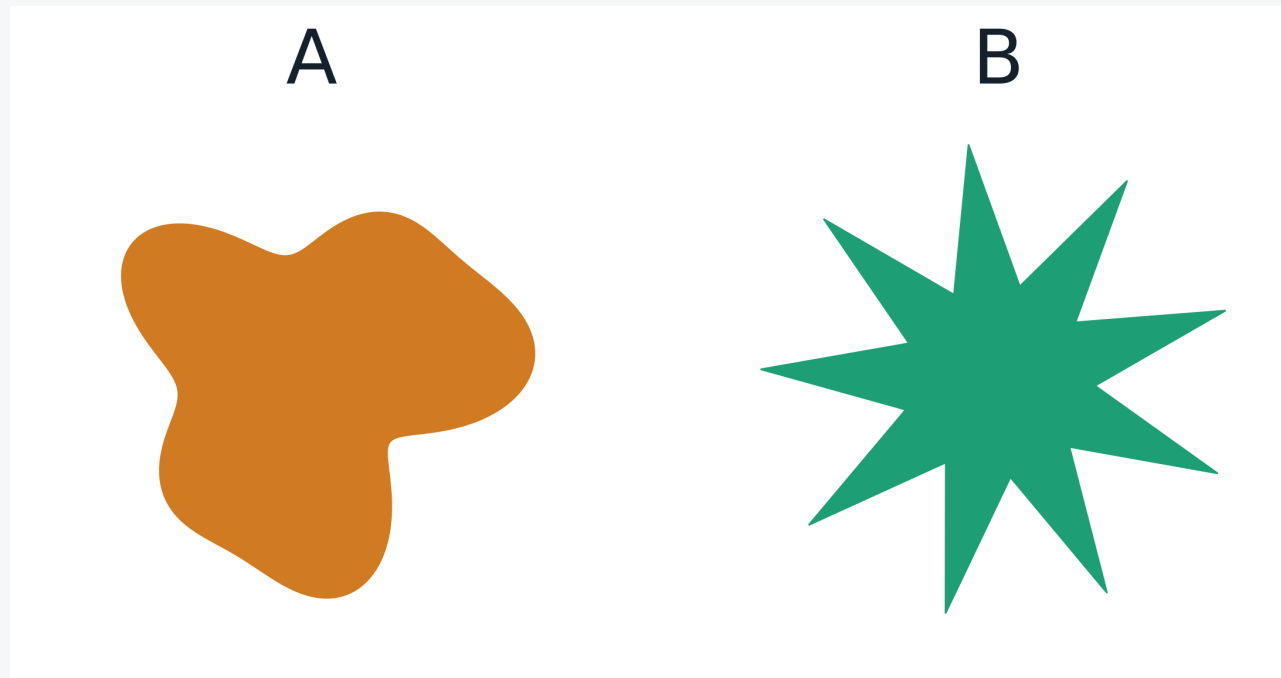
## **multisensory integration**

the brain binds signals from different senses into one event.  
The mechanism behind crossmodal effects

## **multimodal (HCI)**

a system offers several input and output channels: speech plus gesture, display plus vibration

# Which one is “bouba”, which one is “kiki”?



- Which name belongs to which shape?
- A crossmodal **correspondence**: the shape of a sound matched to a shape you see
- A = bouba, B = kiki. About 95 % choose this way, across languages and cultures
  - rounded vowels and soft consonants match rounded forms, sharp consonants match spiky forms
- The same principle lets a designer pair a sound, a haptic pulse and a shape so they feel like one object

# Ba? Ga? Da? The McGurk effect: lips change what you hear



What syllable did he say?

- Pairing in the video: audio ba + lips fa.  
Most people hear **fa**
  - lips touch the teeth → the eyes report an f
  - the sound alone is a clear ba → the ears report a b
  - **the eyes win:** for place of articulation, vision is the more precise sense
- Happens automatically. Knowing the trick does not switch it off
- Not universal: language, age and musical training change the size

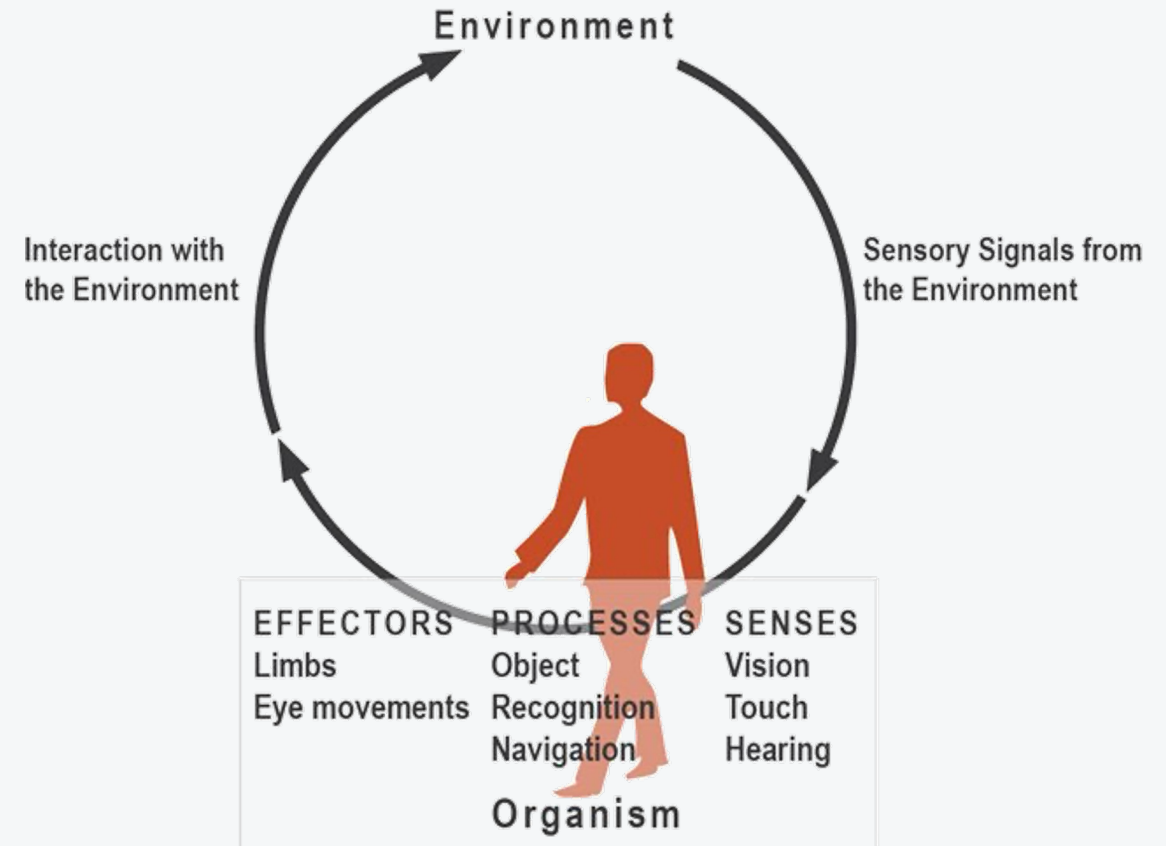
# The human senses, grouped by system, with XR status

| System                  | Sense                        | Stimulus                          | Receptor                        | Consumer XR                     | Research                       |
|-------------------------|------------------------------|-----------------------------------|---------------------------------|---------------------------------|--------------------------------|
| <b>Vision</b>           | sight                        | light                             | retina (rods, cones)            | yes                             | mature                         |
| <b>Audition</b>         | hearing                      | pressure waves                    | cochlea                         | yes, spatial audio              | mature                         |
| <b>Somato-sensation</b> | touch (cutaneous)            | skin deformation, vibration       | 4 mechanoreceptor types         | partial, vibration              | mature                         |
|                         | thermoreception              | skin temperature                  | warm and cold fibres            | no                              | active                         |
|                         | nociception                  | tissue damage                     | free nerve endings              | no                              | avoided                        |
|                         | proprioception / kinesthesia | muscle length, joint angle, force | spindles, tendon organs, joints | partial, controller rumble only | active: grounded devices, EMS* |
| <b>Vestibular</b>       | balance, self-motion         | head rotation, acceleration       | canals, otoliths                | no                              | active: platforms, GVS**       |
| <b>Chemical senses</b>  | olfaction                    | volatile molecules                | ~400 receptor types             | no (first products)             | active                         |
|                         | gustation                    | dissolved molecules               | taste buds                      | no                              | early                          |

\*EMS = electrical muscle stimulation. \*\*GVS = galvanic vestibular stimulation

# Perception is a loop, and XR sits inside it

- Perception is a loop, not a snapshot: we act in order to sense
- Turning the head disambiguates a sound. Moving the hand reveals a texture
- **XR sits inside the loop: tracking reads the user, rendering feeds the senses**
- Presence comes from an intact loop. Latency breaks it, resolution does not fix it



# The claim this lecture defends

1. The brain binds the senses into single events and weights each by how much it can be trusted.
2. XR works by feeding that process. The best XR designs use its rules instead of fighting them.

PART 1 OF 6

# How the brain combines senses

what integration is, how much each sense counts, which sense wins

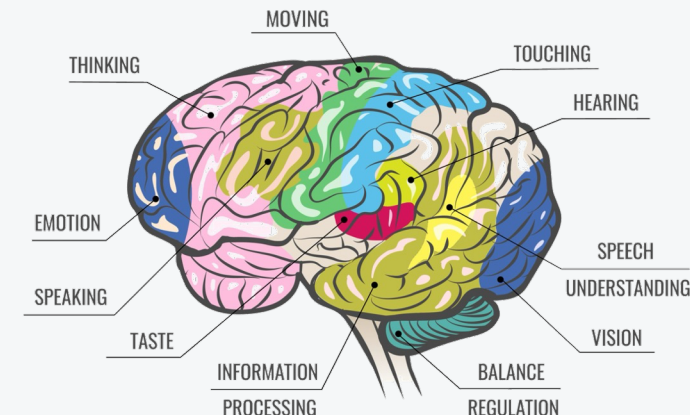
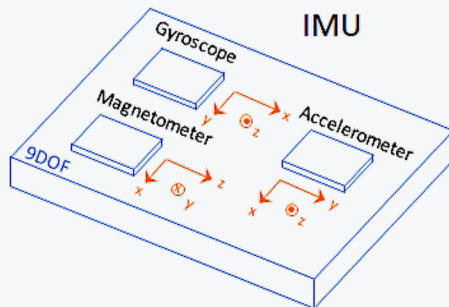
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Every headset relies on this without saying so. After this part you can predict which sense will win any conflict you design.

# Sensor fusion: your phone does it, your brain did it first

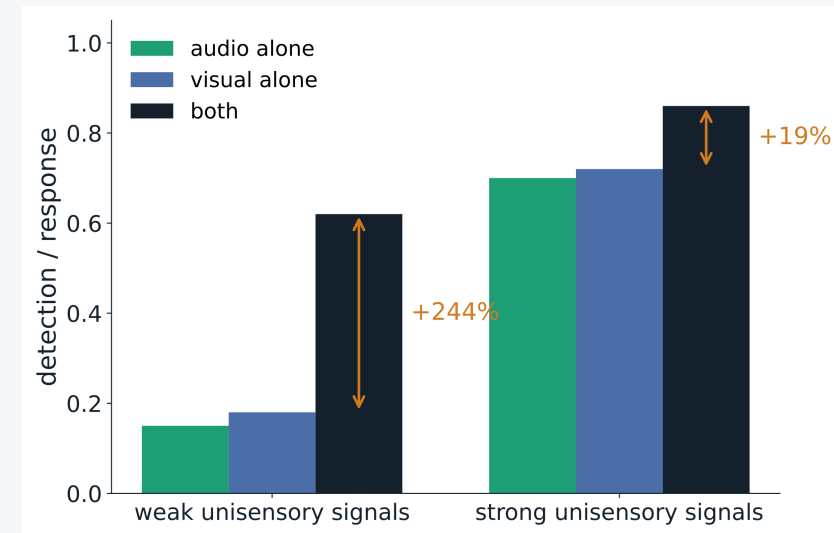
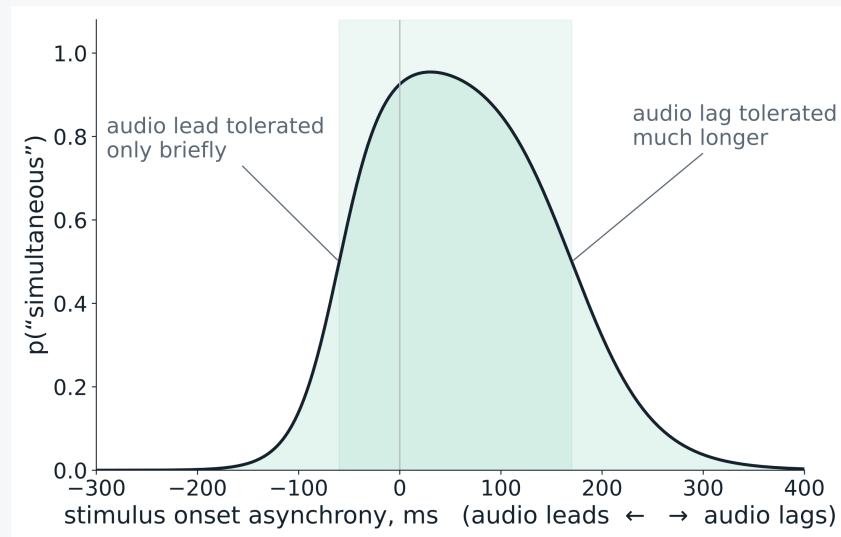
GPS is absolute but noisy, the IMU is precise but drifts, and the phone fuses them.  
Eyes, ears and skin have the same problem and the same solution.

| The problem                     | Your phone                           | Your brain                       | Next slide |
|---------------------------------|--------------------------------------|----------------------------------|------------|
| Which readings belong together? | timestamps, sensor IDs               | coincidence in space and time    | binding    |
| How much to trust each?         | covariance $\rightarrow$ Kalman gain | reliability $\rightarrow$ weight | weighting  |
| What if they disagree?          | gates what it cannot reconcile       | fuses anyway                     | who wins   |

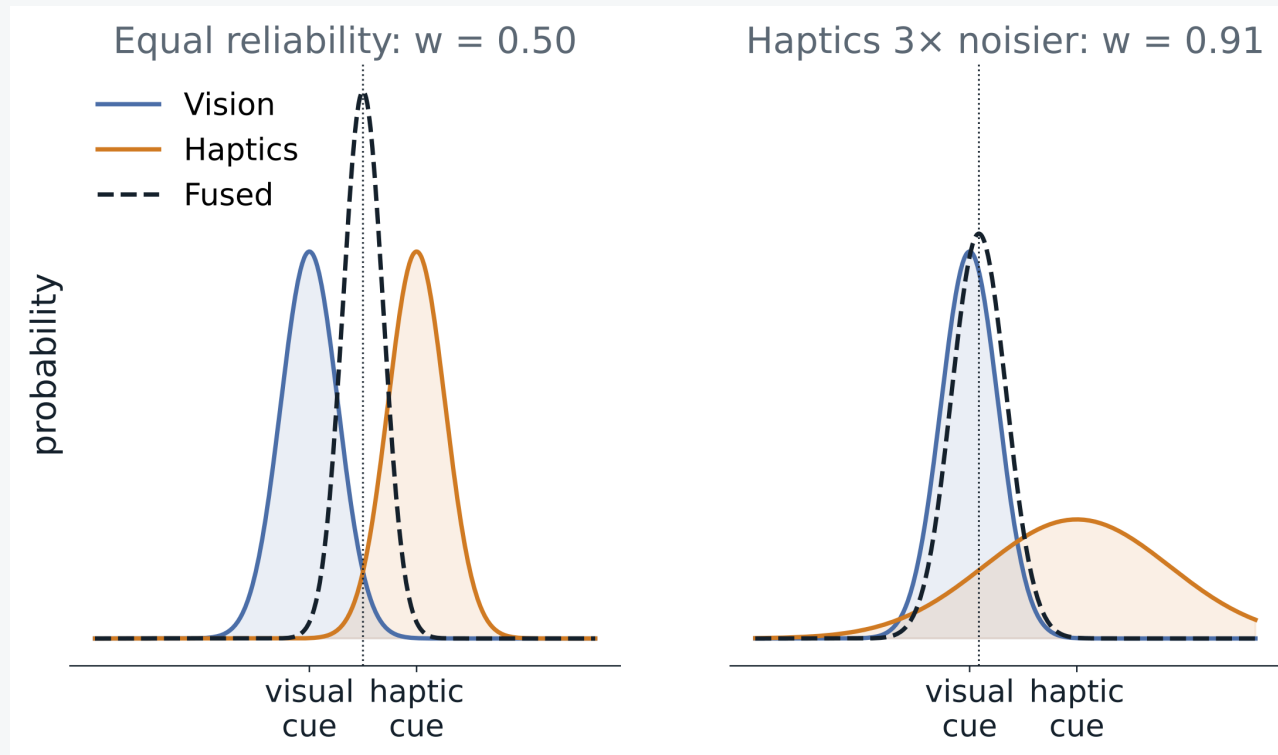


# Integration = binding signals that coincide in space and time

- Spatial rule: signals from the same place get bound
- Temporal rule: signals within about 100 ms get bound. Audio lag is forgiven, audio lead is not
- Inverse effectiveness: the gain is largest when the single signals are weak
- What you get: less noise, faster detection, disambiguation, one object instead of two

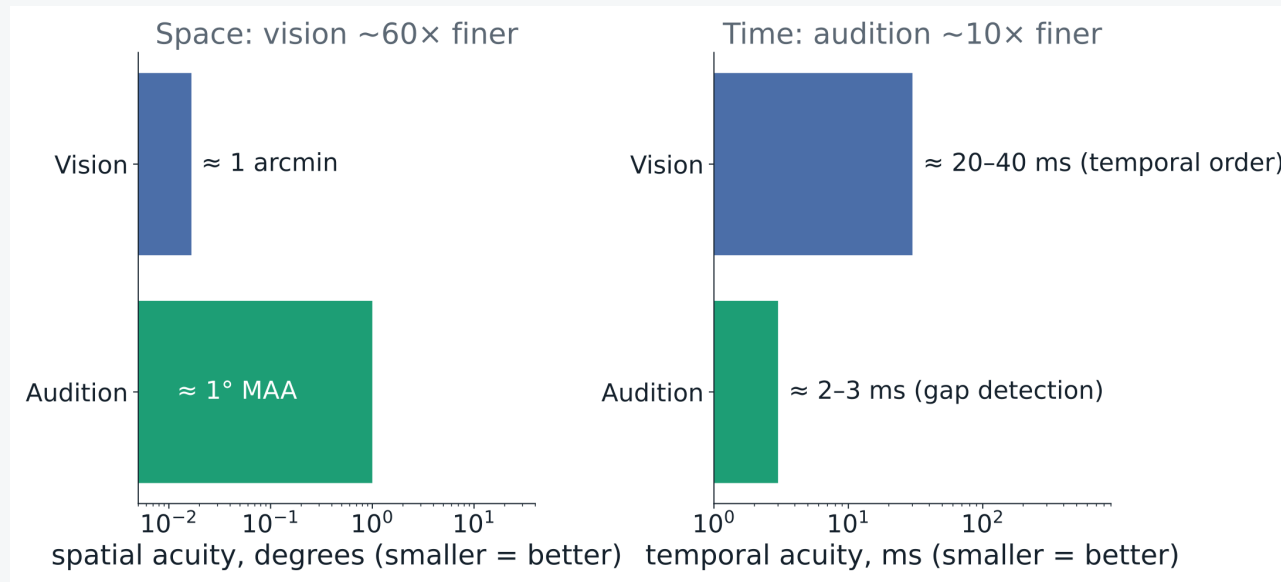


# How much does each sense count? Bayesian cue combination



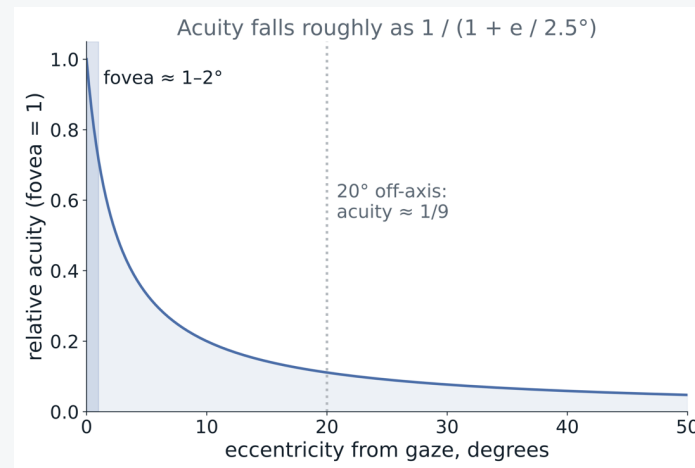
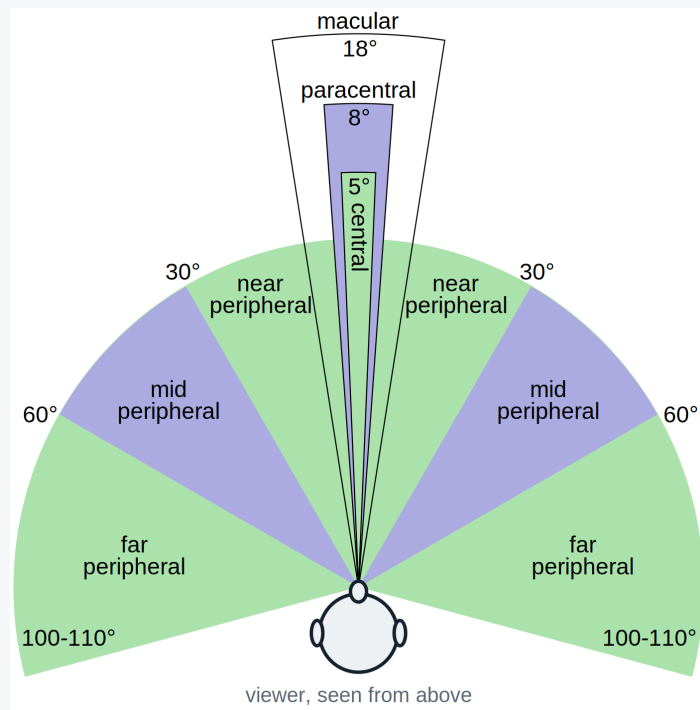
- Each sense delivers an estimate with uncertainty: a distribution, not a single value
- Optimal rule: weight each estimate by  $1 / \text{variance}$ . Reliability means how narrow its distribution is
  - the fused estimate is more precise than either input
- Add noise to one channel  $\rightarrow$  its weight drops, the other takes over (right panel)
- Same law for every pairing of senses

# The rule: the sense that is more precise for the judgement wins



- Space → vision:  $\approx 1$  arcmin ( $1/60^\circ$ ) vs  $\approx 1^\circ$  for hearing
- Time → audition:  $\approx 2-3$  ms vs  $\approx 20-40$  ms for vision
- Not a rule but a consequence of reliability weighting: ventriloquism is space, the flash illusion is time
- Design consequence: the channel you render precisely will dominate that judgement

# Vision (briefly): wide, but sharp and stereoscopic only in the middle



- About 200° wide, stereo only in the 120° both eyes share, detail only in the central 5 to 8°. A headset gives about 110°
- Periphery: coarse but motion-sensitive. Flow alone createsvection
- Depth cues: stereo, parallax, vergence, accommodation

PART 2 OF 6

# Spatial hearing

Coarse in space, exquisite in time, cheap to render

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The cheapest presence in the stack. After this part you can explain what head tracking, HRTF and reverb each add, and in which order to buy them.

# Where is the sound? You already know, without looking



- You locate a voice or a footstep to within a few degrees, eyes closed, without trying
- The input is only two ear signals. No picture of the room, no lens, no retina
- Some directions are easy, others are hard. XR audio has to get all of them right

With only two ears, what could the brain possibly measure to get a direction?

# Two ears, two cues: time and level

- A Interaural Level Difference (ILD):  
the head shadows the far ear, only above about 1.5 kHz. Lower frequencies bend around it
- B Interaural Time Difference (ITD):  
the far ear hears it later, up to 650  $\mu\text{s}$ , resolved to about 10  $\mu\text{s}$
- Duplex theory (Rayleigh): time cues below  $\sim 1.5$  kHz, level cues above
- Together they give one number, the lateral angle. Front/back and elevation need more

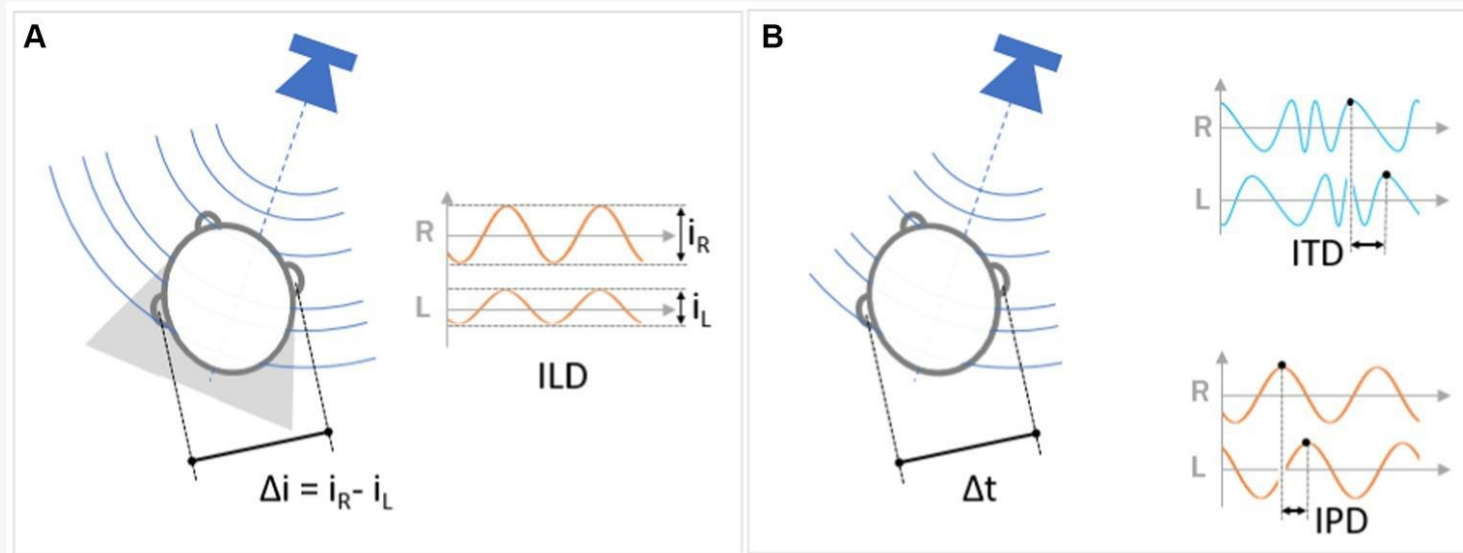


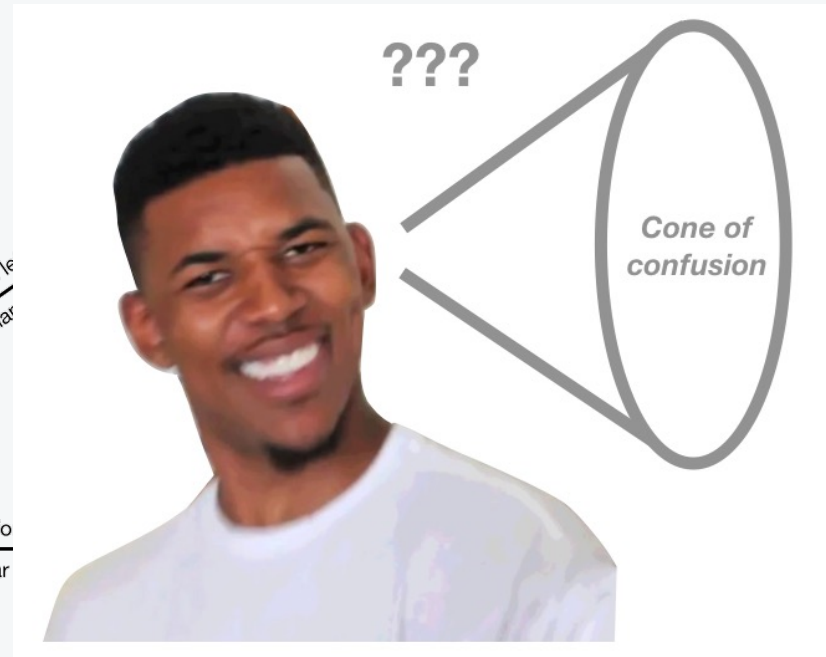
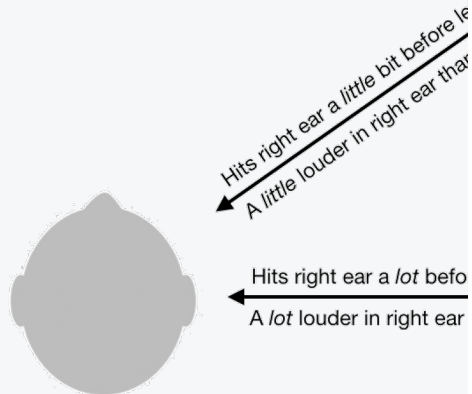
Fig.: Carlini, Bordeau & Ambard, "Auditory localization: a comprehensive practical review", *Front. Psychol.* 15:1408073 (2024), fig. 2, (A) level difference from the head shadow, (B) time and phase difference.. Rayleigh (1907). Measured KEMAR curves in the appendix.

# Lateral angle only → the cone of confusion

- Different points can provide the same ITD and near-identical ILD: front and back, up and down, all on one ring
- Resolved by action and anatomy
  - turn the head: cues change differently for front vs back
  - the pinna filters each direction differently

How to revolve this issue?

Disambiguated

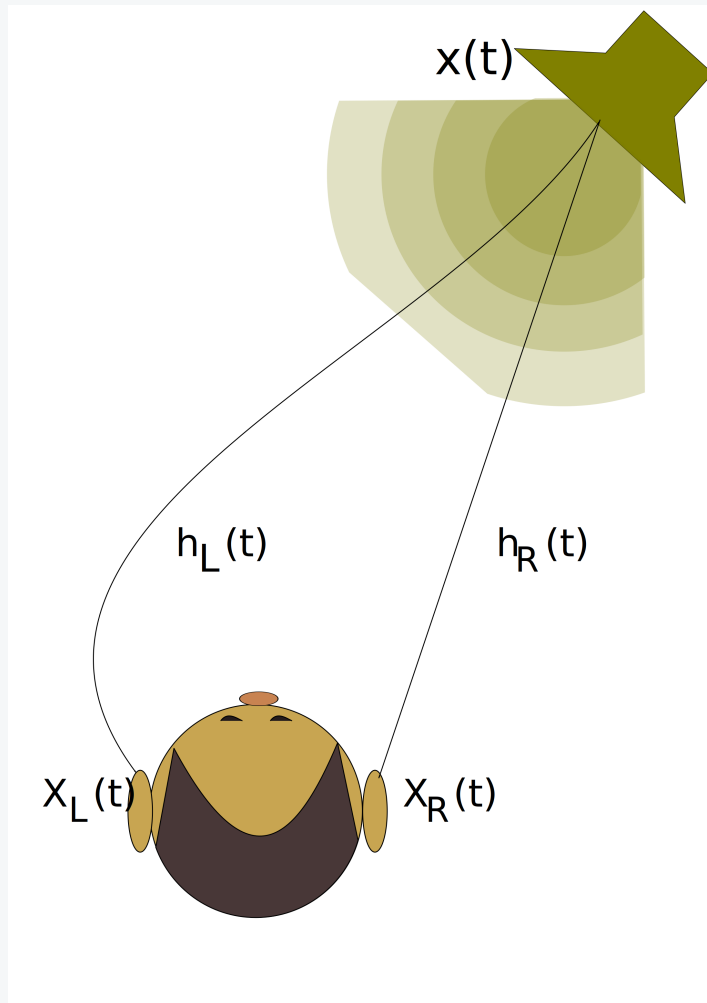


Intensity

Hits right ear a little bit before left ear  
A little louder in right ear than left ear

Hits right ear a little bit before left ear  
A little louder in right ear than left ear

# HRTF: the pinna writes elevation into the spectrum



- Head-related transfer function: for every direction, one filter per ear,  $h_L(t)$  and  $h_R(t)$
- The pinna's folds add notches at 6 to 10 kHz that move with elevation. The brain has learned your notch map
- Rendering = filter (convolve) each source with the HRTF of its direction
- This turns left/right into a full 3-D direction

# Generic or individual HRTF? Every head is different



- Generic: a dummy head with microphones in its ears  
Individual: measured on you or picked by listening test
- Pinnae differ between people. Even a dummy head's two ears give different spectra above 6 kHz
- Jenny & Reuter 2020, 39 listeners in VR: own HRTF localizes better + changing subjective experience
- Measuring HRTF is expensive!

PART 3 OF 6

# Haptics

fast in time, coarse in space, hard in hardware

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The channel consumer XR left half-built. After this part you can say why a controller buzzes but nothing pushes back, and match an actuator to a receptor.

# What you feel when you pick up a mug

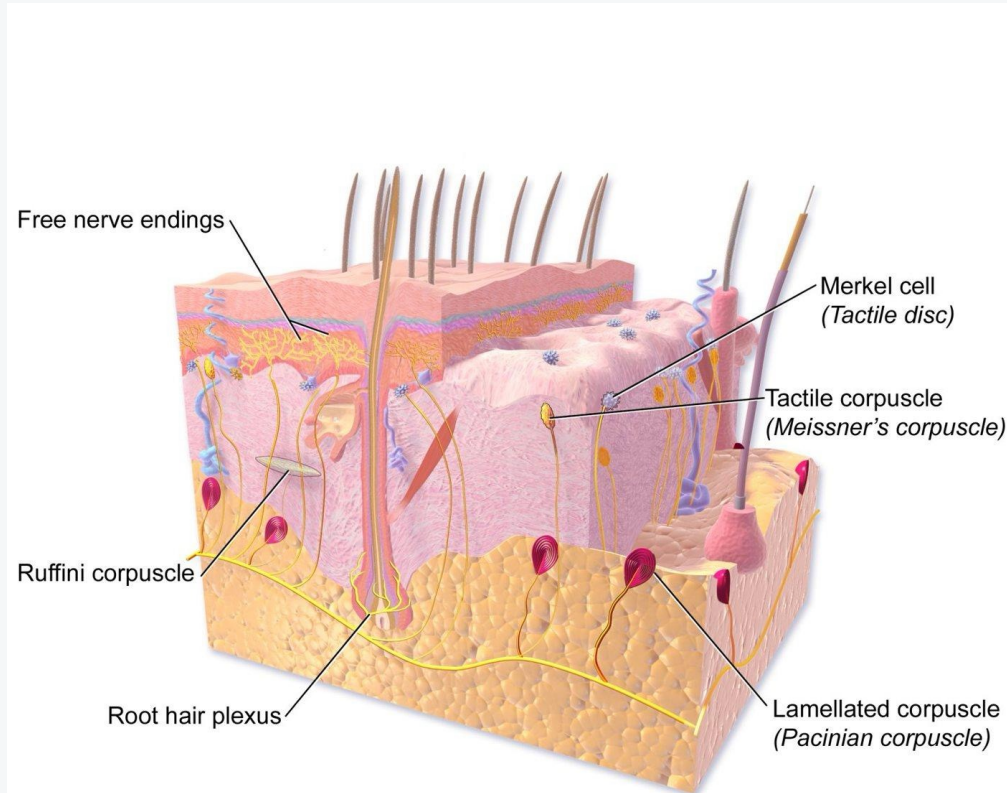


- (At least) five judgements in one grasp
  - Weight
  - Texture
  - Temperature
  - Slip
  - Stiffness

Which of the five could a controller give you today?

- The next slides split the five into two systems and show which one XR can render

# Two systems: skin and muscle



**Tactile Receptors in the Skin**

- Cutaneous (tactile): mechanoreceptors in the skin
  - texture, vibration, pressure, slip, edges
  - glabrous skin (palms, soles, lips) has fine acuity. Hairy skin has follicle receptors and slow affective touch
- Kinesthetic: muscle spindles, Golgi tendon organs, joint receptors
  - force, weight, stiffness, limb position
- Together they are haptics. Gripping a mug uses both
- XR renders the first cheaply (vibration) and the second barely (force)

# Touch receptors and what can drive them

Most concentrated in glabrous skin

| Receptor (channel)                          | Adapts* | Best frequency           | Field size     | Feels like                        | Driven in XR by                                                                               |
|---------------------------------------------|---------|--------------------------|----------------|-----------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Pacinian (PC)</b>                        | rapid   | 40–500 Hz, peak ≈ 250 Hz | large, deep    | vibration, buzz, “contact” pulses | LRA (linear resonant actuator), voice coil, ultrasound mid-air, ERM (Eccentric Rotating Mass) |
| <b>Meissner (RA I)</b>                      | rapid   | 3–40 Hz                  | small, shallow | flutter, slip, tapping            | low-frequency voice coils                                                                     |
| <b>Merkel (SA I)</b>                        | slow    | < 10 Hz, static          | small          | pressure, edges, texture, form    | pin arrays, pneumatic pads (research)                                                         |
| <b>Ruffini (SA II)</b>                      | slow    | static                   | large          | skin stretch, hand shape          | skin-stretch tactors (research)                                                               |
| <b>Thermoreceptors</b>                      | slow    | -                        | -              | warm / cool                       | Peltier elements, chemical haptics                                                            |
| <b>Spindles &amp; tendons (kinesthetic)</b> | -       | -                        | -              | force, weight, stiffness          | grounded arms, exoskeleton gloves, EMS*                                                       |

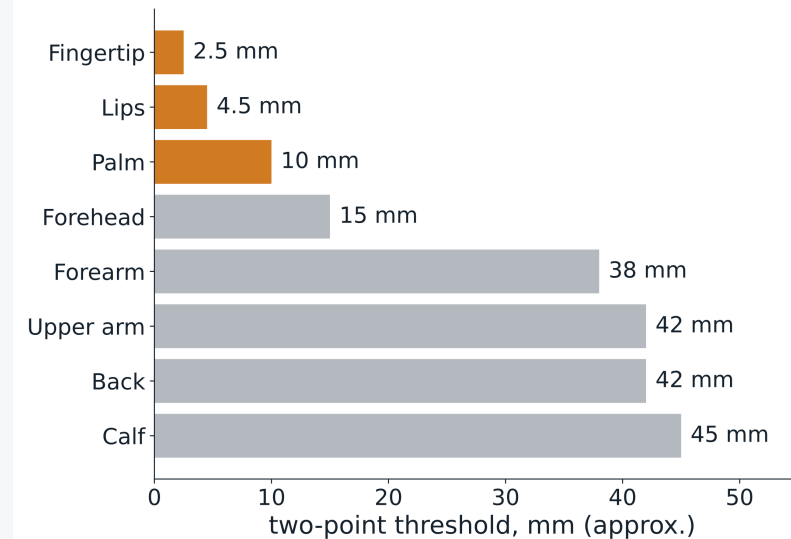
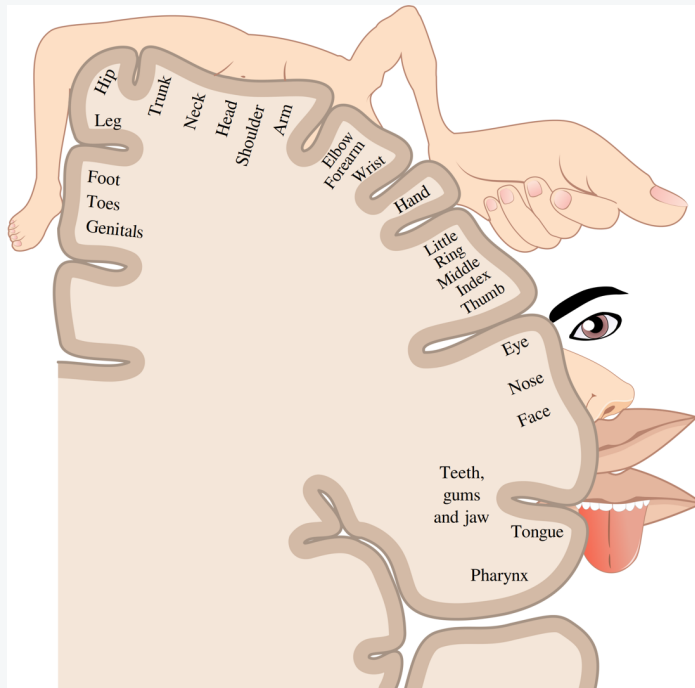
—\*EMS = electrical muscle stimulation.

\*adapting = stops firing under constant pressure. LRA = linear resonant actuator, ERM = eccentric rotating mass. Green = commercial, amber = research, red = hard

Bolanowski et al., *J. Acoust. Soc. Am.* 84:1680 (1988). Johnson, *Curr. Opin. Neurobiol.* 11 (2001).

# Spatial resolution: not uniform across the body

- Left, the sensory homunculus: the body map in the cortex, scaled by acuity, not by body size
  - fingertip  $\approx$  2 mm two-point threshold, back  $\approx$  40 mm
- A vest needs few actuators, a glove needs many tiny ones. Placement beats actuator quality
- Temporal side is the opposite: skin resolves  $\sim$ 1 kHz, far faster than vision



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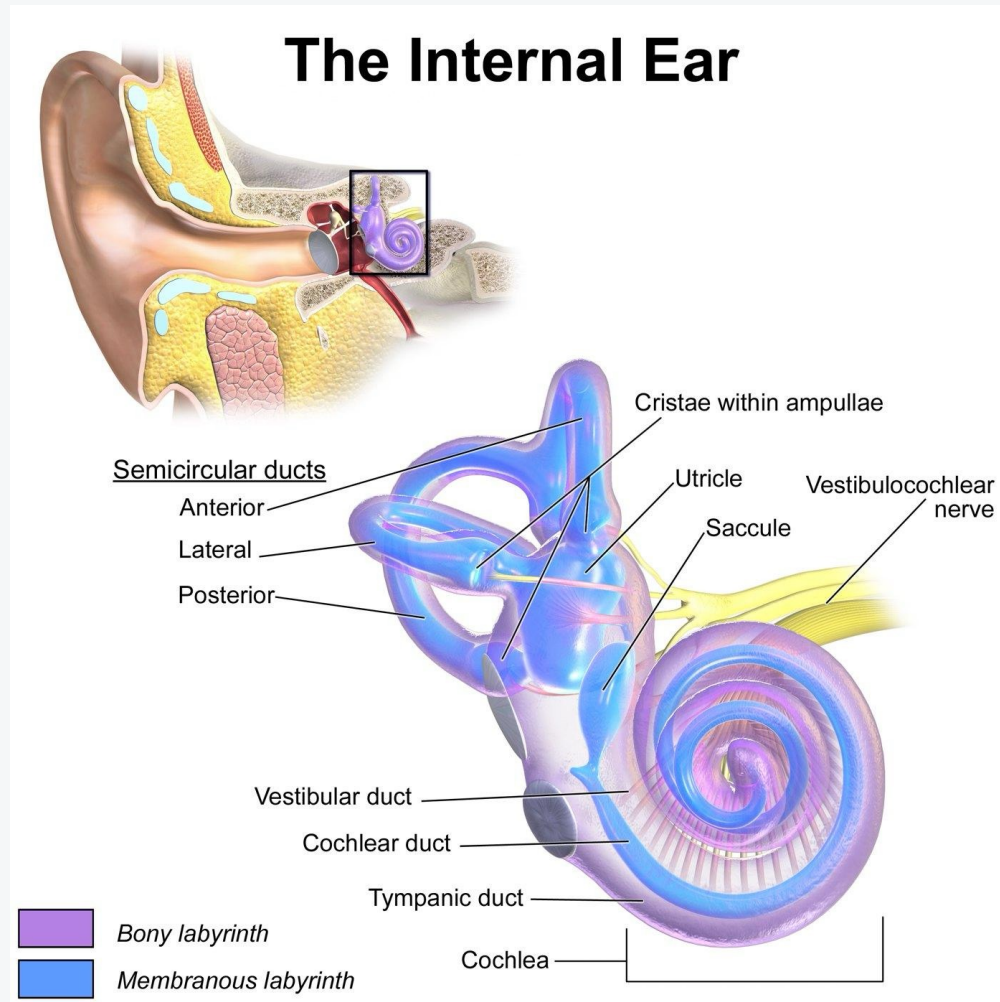
# Vestibular & Proprioception

Balance and body-position: always active, and the only senses we can disturb rather than render

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Everything so far described how perception works. This is where you learn to use it, and check whether your design actually worked.

# The vestibular sense: the inner ear's motion sensors



- Three semicircular canals, one per axis, sense rotation of the head
- Two otolith organs (utricle, saccule) sense linear acceleration and gravity: a lift starting, a tilt
- Reflexes stabilise gaze in ~10 ms and posture before you notice
- Always on, never rendered, and it votes in every self-motion judgement

# Proprioception: knowing where your body is without looking

## Muscle spindles

inside each muscle. Report length and stretch speed → limb position and movement

## Golgi tendon organs

at the tendon. Report tension → how hard you push or hold

## Joint & skin receptors

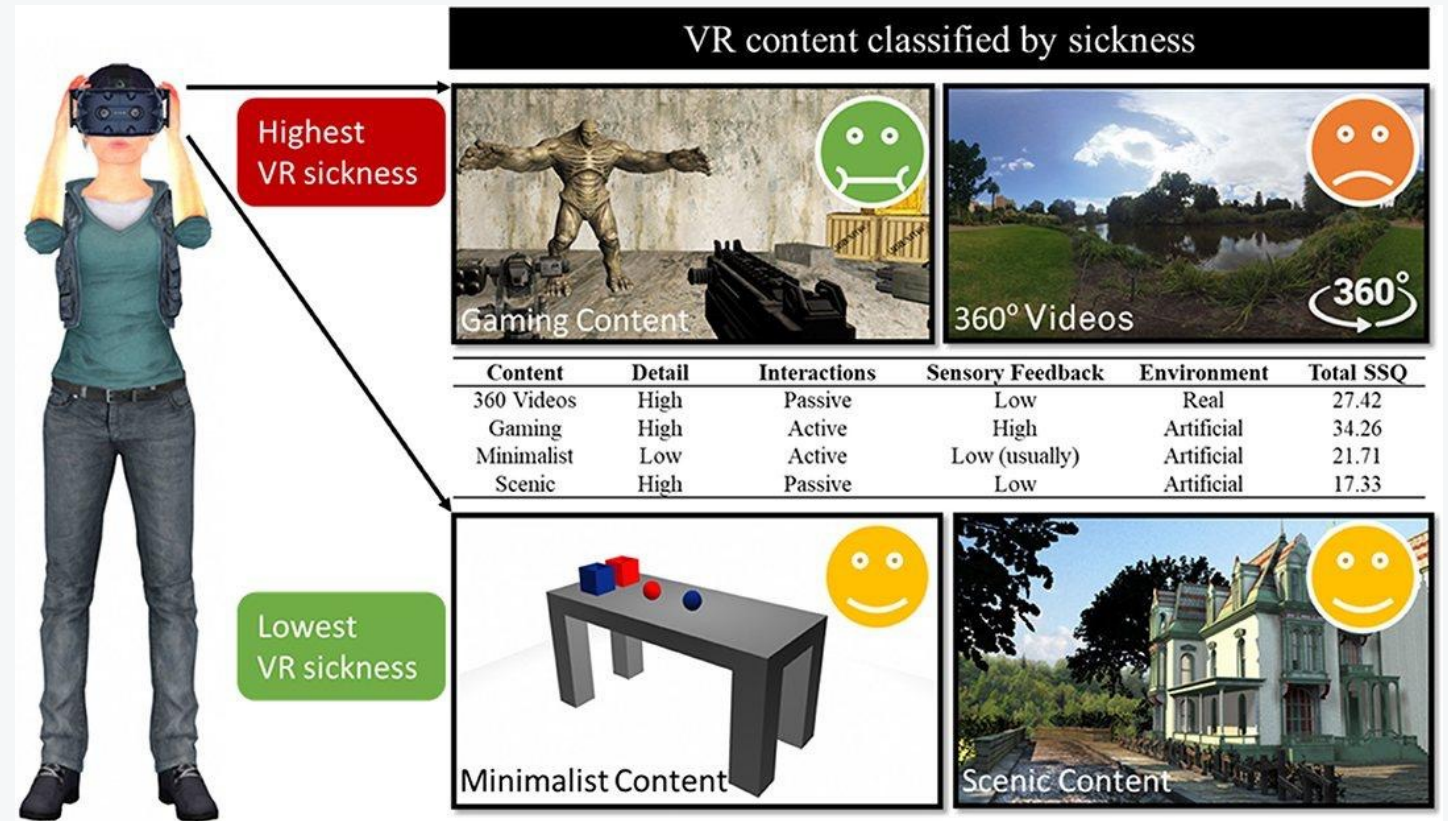
capsules and stretching skin. Report joint angle and its limits

- In XR, hand tracking reads it and nothing feeds it. When seen and felt hand disagree, vision usually wins (the lever we use later)

Eyes closed: touch your nose with your index finger. That is proprioception.

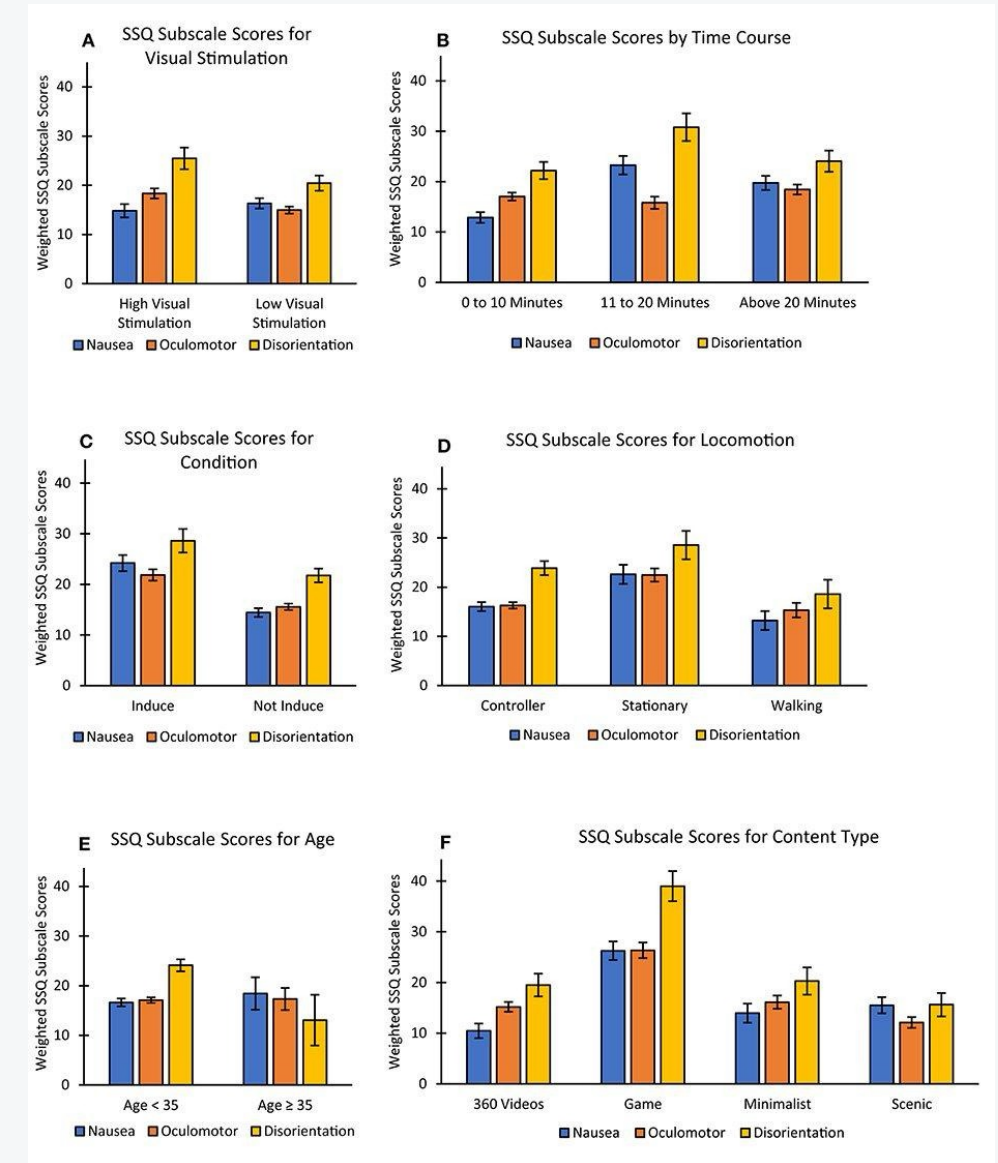
# Cybersickness: the eyes say “moving”, the canals say “still”

- Sensory-conflict theory: visualvection, vestibular silence. The binding rule finds no partner
  - rival view: postural instability
- Measured with the SSQ (Simulator Sickness Questionnaire): nausea, oculomotor, disorientation
- Worst: 360° video and fast games. Mildest: scenic and minimalist content (figure)



# What actually reduces it

- Locomotion: walking or teleport beats controller-driven movement (panel D)
- Reduce vection: narrow the FOV during motion, add a stable frame
- Latency < 20 ms motion-to-photon,  $\geq 90$  Hz
- Give the canals **something**: seat vibration, galvanic vestibular stimulation (GVS), motion base
- Habituation over sessions (panel B), but you lose the users who quit first



# Galvanic Vestibular Stimulation (GVS)



- Two electrodes behind the ears, a few milliamps. The vestibular nerve fires as if the head tilted
- Result: felt self-motion and a real lean. The video shows a walker veering under 5 mA
- XR research: a vestibular cue to match visual acceleration (Maeda et al. 2005)
- Our only levers on this sense: motion platforms, seat shakers, GVS, redirected walking

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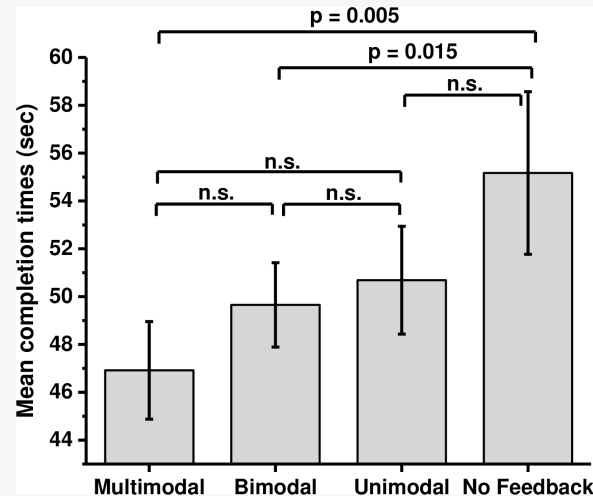
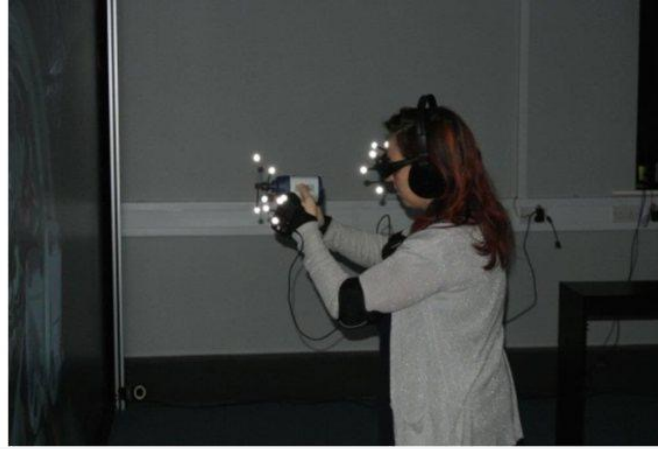
# Designing with integration

Perception isn't fixed. You can design for it, and measure the result.

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Where the rules become tools. After this part you can stage a sensory disagreement deliberately and know its limit.

# Does adding a sense help the task? Yes, measurably



- Cooper et al. 2018: VR wheel-change task, audio, vibration and visual contact cues
  - all combinations tested
- All three cues about 47 s, no cue about 55 s. Biggest jump: any versus none
- Presence ratings followed the same order
- Cheap, congruent cues beat fidelity in one channel

# Three levers: when vision and a body sense disagree, vision wins

XR can stage that disagreement on purpose.

## 1. Ownership

correlate seen and felt touch → the brain adopts a virtual body as **me**

## 2. Redirection

vision outweighs proprioception → warp the seen hand or path. One prop, many virtual objects

## 3. Pseudo-haptics

vision alone reports weight, stiffness, friction → change the control–display ratio, no hardware

# Ownership in VR: a whole body, three metres away



- Pomés & Slater 2013: vibrotactile vest + the same touch seen on an avatar in front of the participant
- Synchronous touch → ownership of the distant body, and the felt self drifts toward it
  - asynchrony above about 300 ms breaks it, first-person view strengthens it
- Lever one: seen and felt touch outvote proprioception. The basis of avatars and social XR

# Redirection: vision wins in space, so warp the hand



- One physical cube stands in for many virtual ones. The seen hand path is warped so the real hand meets the prop
- Works because proprioception loses to vision within a measurable tolerance
- Redirected walking: the same trick bends a straight path into a circle. Haptics bought with perception, not actuators

# Pseudo-haptics: feeling force through vision alone

- Visual feedback that disagrees with the hand's motion. The brain reads the mismatch as a force
- The knob is the control-display ratio: hand moves 10 cm, virtual hand 7 cm → heavier, stiffer (Dominjon 2005, Samad 2019)
- Also stiffness and texture: a cursor that slows over a patch makes it feel harder (Li et al. 2016)
- Cheap, measurable, stacks with real vibration. Fails once the mismatch is obvious



# Three design rules that follow from the principles

## **Congruence beats fidelity**

cues from the same location bind into one percept, even at low fidelity (spatial rule)

## **Respect the binding window**

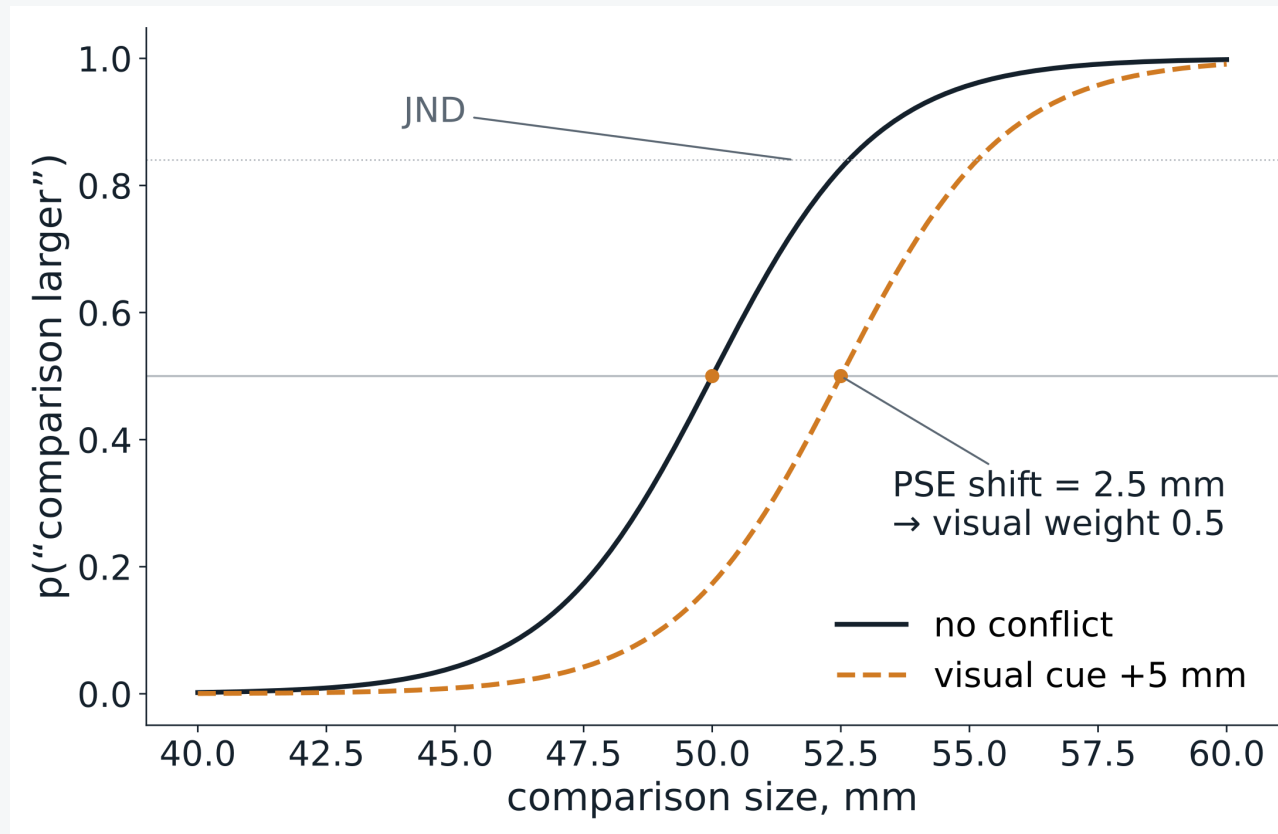
about 100 ms audiovisual, about 50 ms visual-haptic. Audio lag is tolerated more than audio lead (temporal rule)

## **Add where the channel is weak**

a second sense helps most where the first is degraded: touch in the periphery, sound behind you (inverse effectiveness)

Rule of thumb for budgets: spend on synchrony and placement before spending on resolution.

# Measure integration with psychophysics, not only questionnaires



- Psychophysics can measure how strongly each cue is weighted
- Cue-conflict method: offset one cue, and find the point where the judgement is equally likely to go either way, the point of subjective equality (PSE)
  - $\text{PSE shift} \div \text{offset} = \text{weight}$
  - Slope = just-noticeable difference (JND) = reliability of the channel

# Rendering budget per modality

| Modality                   | Needs tracked                                                                | Update rate, latency target                                     | Typical actuator                      |
|----------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------|
| <b>Visual</b>              | head 6-DoF = position and rotation (required). Eyes for foveation (optional) | 90–120 Hz, motion-to-photon (head move → updated image) < 20 ms | HMD displays                          |
| <b>Auditory</b>            | head rotation (required). Position only for near sources and room            | 48 kHz, 5–20 ms buffers, HRTF per source                        | headphones, open-ear speakers         |
| <b>Vibrotactile</b>        | hands / contact events (standard)                                            | 1 kHz update, pulse at contact ± 50 ms of the visual            | LRA, voice coil, vest arrays          |
| <b>Force / kinesthetic</b> | hand pose (standard). Body (optional)                                        | 1 kHz servo loop                                                | grounded arm, exoskeleton glove, EMS* |
| <b>Thermal</b>             | contact events                                                               | seconds, so pre-trigger before contact                          | Peltier elements                      |
| <b>Olfactory</b>           | scene events                                                                 | 0.1 to 2 s onset, so pre-trigger, then clear                    | micro-pumps, fans, heated wax         |
| <b>Vestibular</b>          | head motion, locomotion intent                                               | match visual acceleration onset                                 | motion base, GVS**, seat shaker       |
| <b>Physiology (input)</b>  | heart rate, skin conductance (optional)                                      | for adaptation and evaluation, not rendering                    |                                       |

\*EMS = electrical muscle stimulation. \*\*GVS = galvanic vestibular stimulation. Required, standard, optional = what a 2026 consumer headset tracks by default

Own summary. Abrash (2014). Begault, Wenzel & Anderson (2001). Lindau (2009). Massie & Salisbury (1994). Vroomen & Keetels (2010).

PART 6 OF 6

# Five things an extra sense can do

organise related work by function, not by modality

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A way to sort related work. After this part you can place your own project in one of five functions.

# Five topics to sort (almost) any multisensory XR paper

## Guide

the cue steers  
attention or the  
body

## Deceive

one sense overrules  
another, to save  
hardware

## Deepen

presence rather than  
performance

## Transform

the multisensory  
context rewrites a  
real object

## Substitute

a channel replaces  
one the user does  
not have

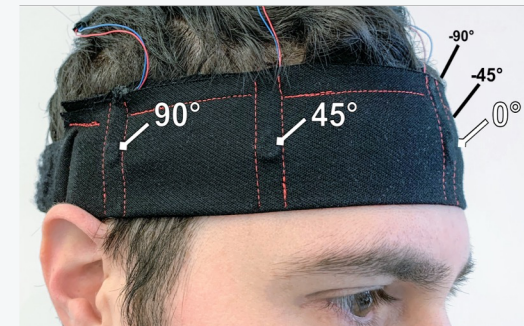
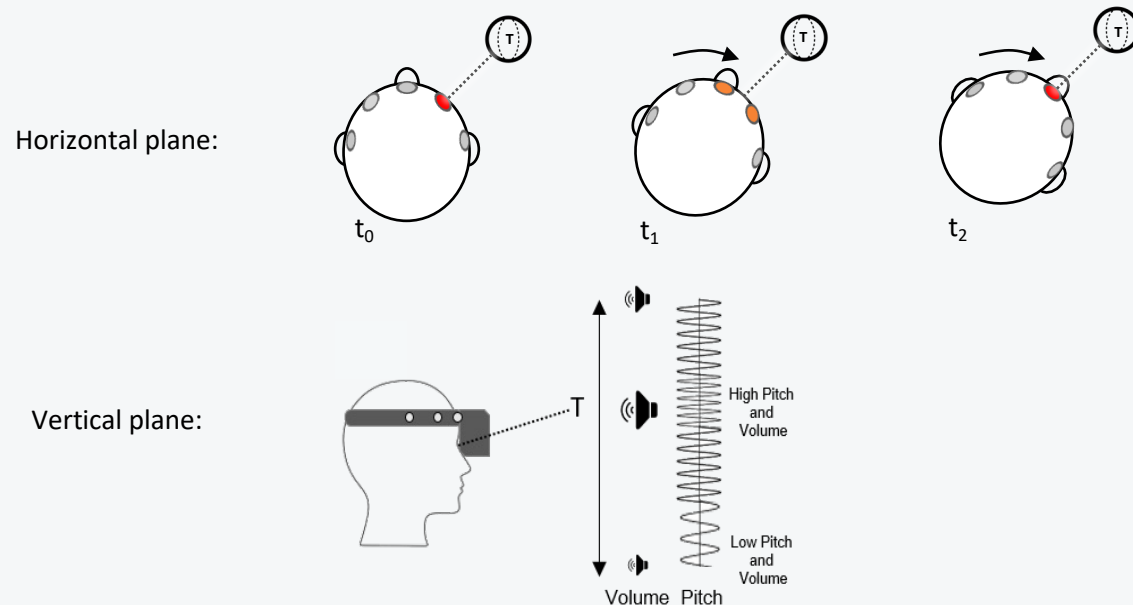
Deceive you have seen (redirection, pseudo-haptics). One example for each of the other four follows. at the end, the question is which verb is yours.

# Guide: the stimuli steer attention

## *Comparing Non-Visual and Visual Guidance Methods for Narrow FoV Augmented Reality Displays*

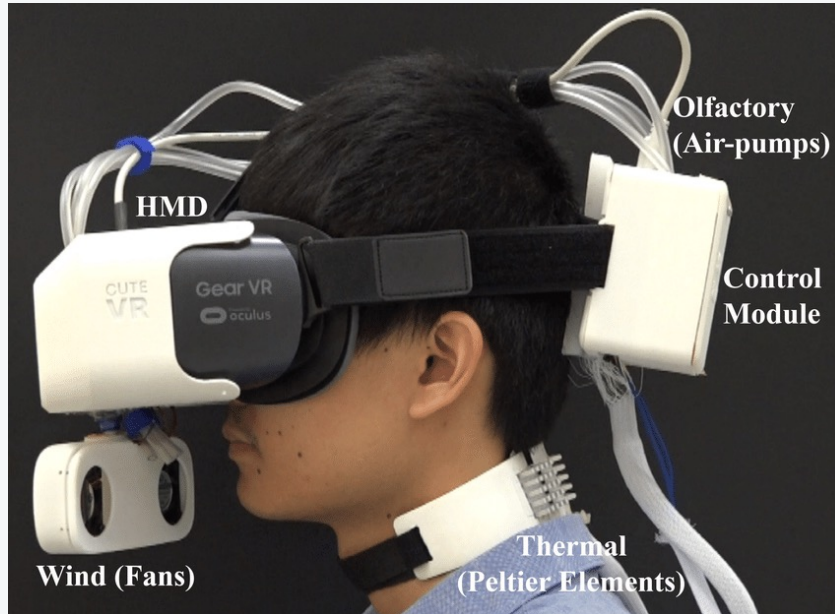
### Out-of-view targets, located with other senses

- Tactile headband, five vibration motors around the temples and forehead, plus auditory cues
  - Horizontal direction: vibration motors positioned around the head indicate direction
  - Elevation: pitch and volume rise and fall with target height
  - Depth: vibrotactile pulses on the head
- Found targets as accurately as visual guidance, but notably slower

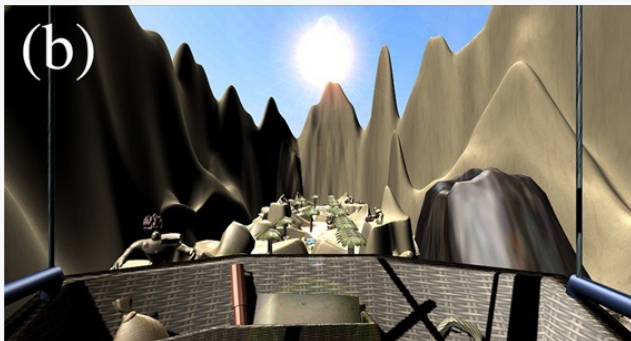


# Deepen: presence rather than performance

*Season Traveller: Multisensory Narration for Enhancing the Virtual Reality Experience*



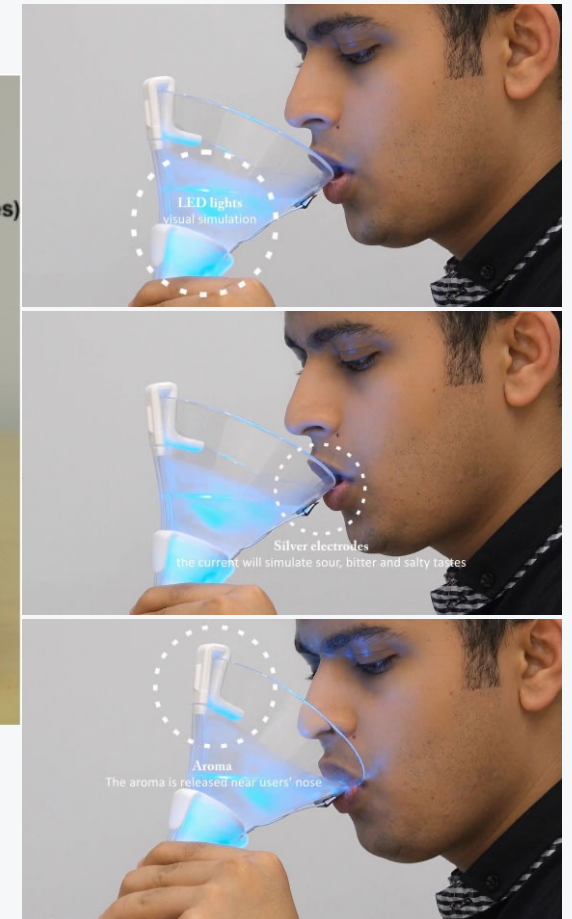
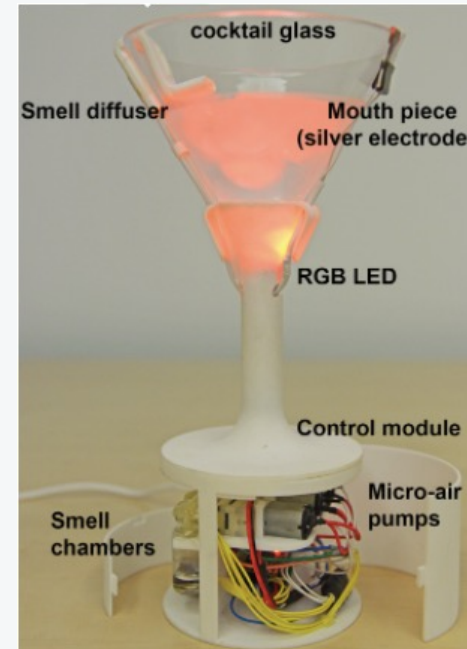
- Four seasons in VR with wind, heat, cold and scent added to audio-visuals
- Presence rose with each added ambient channel measurably
- Presence is the deliverable here, so measure it as such



# Transform: alter the percept of a real thing

*Vocktail: A Virtual Cocktail for Pairing Digital Taste, Smell, and Color Sensations*

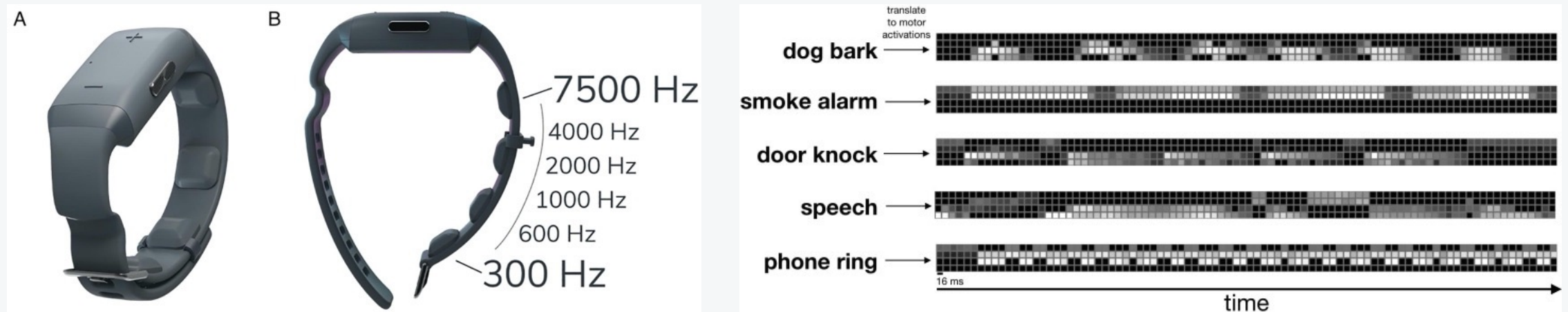
- Electrodes on the rim stimulate the tongue  
→ salty or sour
- Scent on the surface, LED changes the water's colour
- Combined stimuli (electric, scent, visual colour) gave richer flavours than any single one
- Function: the real object stays, the multisensory context rewrites it



# Substitute: replace a channel the user does not have

## *Deciphering Sounds Through Patterns of Vibration on the Skin*

- A wristband converts sound into vibration patterns on the skin
- Deaf and hard-of-hearing users learned to identify different sounds from the pattern alone, and to tell similar-sounding words apart with practice
- The brain reads whichever channel carries the statistics



# What to take home

- Integration binds by space and time and weights each sense by its reliability
  - space → vision, time → audition, and every illusion follows
- Binaural hearing is the cheap win: ITD / ILD, head tracking, then HRTF
- Haptics: vibration solved, force hard, so borrow from perception
- The unrenderable senses set the limits: latency, vection, conflict
- Measure with psychophysics. Design for congruence, not fidelity

**feed the process,  
don't fight it**

**Guide**  
**Deceive**  
**Deepen**  
**Transform**  
**Substitute**

Which one is your project?

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# Glossary

|                          |                                                                                            |                       |                                                                                       |
|--------------------------|--------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------|
| Multisensory integration | the brain binds signals from several senses into one percept and combines them             | Presence              | feeling of being there and reacting as if it were real                                |
| <b>Reliability</b>       | how narrow a sense's estimate is (1 / variance). Sets its weight                           | Vection               | felt self-motion caused by visual flow alone                                          |
| <b>Psychophysics</b>     | measuring perception with controlled stimuli and simple judgements                         | SSQ                   | Simulator Sickness Questionnaire: nausea, oculomotor, disorientation                  |
| <b>PSE / JND</b>         | point of subjective equality and just-noticeable difference, read off a psychometric curve | Motion-to-photon      | delay from a head movement to the updated image                                       |
| <b>ITD / ILD</b>         | interaural time and level difference, the two binaural cues                                | 6-DoF                 | position and rotation tracked, three axes each                                        |
| <b>HRTF</b>              | head-related transfer function: per-direction, per-ear filter of the head and pinna        | Control–display ratio | how far the virtual hand moves per unit of real hand movement                         |
| <b>KEMAR</b>             | standard dummy head with microphones in its ears, used to measure generic HRTFs            | EMS / GVS             | electrical muscle / galvanic vestibular stimulation                                   |
| <b>Externalisation</b>   | hearing a headphone sound outside the head rather than inside                              | LRA / ERM             | linear resonant actuator and eccentric rotating mass, the two common vibration motors |
| <b>Proprioception</b>    | sense of limb position and force from muscles, tendons, joints                             | Glabrous skin         | hairless skin of palms, soles, lips, the high-acuity touch surface                    |